Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/05/2012
PI Name:	Dantzig, Jonathan A Ph.D.		
Project Title:	Modeling Peritectic Microstructure Formation during Directional Solidification in Space and on Earth		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Materials Science Rapid Turn Around Flight Opportunity
Start Date:	06/03/2010	**End Date:**	06/02/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Period of performance changed to 6/3/2010-6/2/2014 (previously 4/6/10-4/5/14) per NSSC (jvp/editor 7/14/10)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10AJ73G		
Performance Goal No.:			
Performance Goal Text:			

Peritectic alloys constitute an important class of industrial alloys. The properties of these alloys depend strongly on their solidification history, and the potential exists to create novel materials with unique properties by controlling the solidification process. Peritectic alloys undergo an invariant reaction, in which a liquid phase l reacts with an existing solid phase alpha to form a second solid phase beta. When solidification occurs at very low growth rate in a high temperature gradient, this reaction can produce very complicated, non-equilibrium microstructures due to the limited diffusion in the solid state, and convection in the liquid.
The European Space Agency (ESA) has an ongoing multi-national, multi-investigator program, called METCOMP, to study peritectic alloys. The proposed research will provide modeling support for one of these METCOMP projects, focusing on directional solidification in Cu-Sn alloys. Experiments performed on Earth are difficult to interpret because

Task Description:	of the possibility of buoyancy-driven convection. Such experiments are necessary, however, to provide the scientific base for interpreting the space experiments. The objective of the proposed research is to develop and use models that do two things: (1) interact with the experimental work so as to improve the experiments themselves; and (2) provide detailed and predictive models to assist in interpretation of the experimental results. In order to accomplish the first objective, separate models for the macroscale transport phenomena and microstructure formation will be developed first. These models will then be merged into a single code that computes the transport and microstructure at the same time. This is feasible because the code will build on an existing code that uses adaptive mesh refinement to allocate computational resources efficiently.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research, if successful, will lead to improved products for peritectic alloys in general, and for Cu-Sn alloys in particular. There is experimental evidence that coupled growth in these systems can lead to novel microstructures with unique properties. However, these structures are difficult to produce and control This research could help resolve this problem, making these materials commercially available.
Task Progress:	Research in the first year examining the onset of solutal buoyancy-driven convection showed that it is possible to achieve convection-free directional solidification of Cu-[14-20]wt% Sn alloys in a terrestrial environment by making the sample sufficiently small in diameter. This was first predicted in calculations performed under this grant, and then evidenced in experiments conducted by our ESA partners at Ecole Polytechnique Fédérale de Lausanne (EPFL). Due to the success of these ground-based experiments in producing composite microstructures on earth, most of the work this year was focused on the prediction of microstructures during directional solidification in a convection-free environment. Two different computational approaches are being pursued. The first approach is the adaptation of Prof Mathis Plapp's code, originally developed for eutectics, for the study of peritectics. This code has been used before at EPFL, but it was limited to artificial parameter sets (reduced spacing, increased speed and temperature gradient). In this report, results are given for simulations using realistic parameter values, corresponding to those used in the experiments described above. The results show that we can indeed produce lamellar microstructures, but that they are rather unstable unless the temperature gradient is very large. This year, the code has been modified and adapted to model realistic values of the experimental parameters. This is work in progress. The second approach is the continued enhancement of a previously developed adaptive phase-field code for binary alloy solidification. In Year 1, the code was extended to incorporate generalized anisotropy functions based on expansion in spherical harmonics. A number of interesting results were obtained while examining the role of the surface energy anisotropy in dendrite orientation selection.
Bibliography Type:	Description: (Last Updated: 04/03/2014)
Abstracts for Journals and Proceedings	Friedli J, Di Napoli P, Rappaz M, Dantzig JA. "Phase-field modeling of the dendrite orientation transition in Al-Zn alloys." To be presented at MCWASP-2012 (Modeling of Casting, Welding and Advanced Solidification Processes 2012), Styria, Austria, June 17-22, 2012. Modeling of Casting, Welding and Advanced Solidification Processes 2012, June 2012. In Press. , Jun-2012